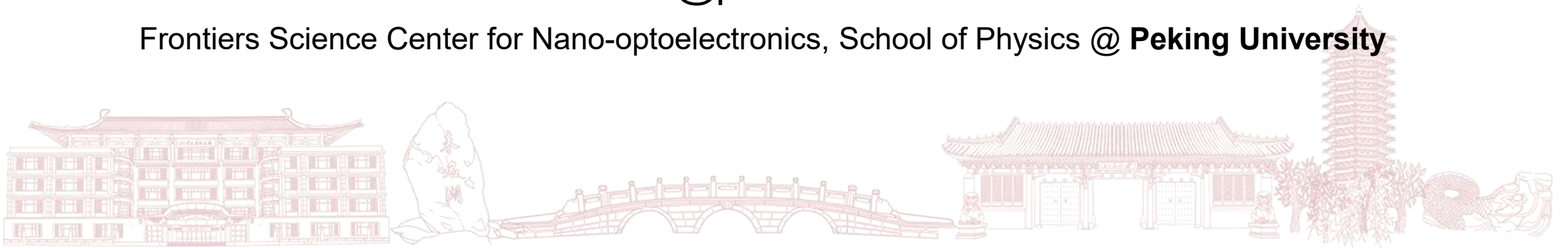


Trapping Ytterbium atoms in optical tweezers

Xiang-Ru Xie

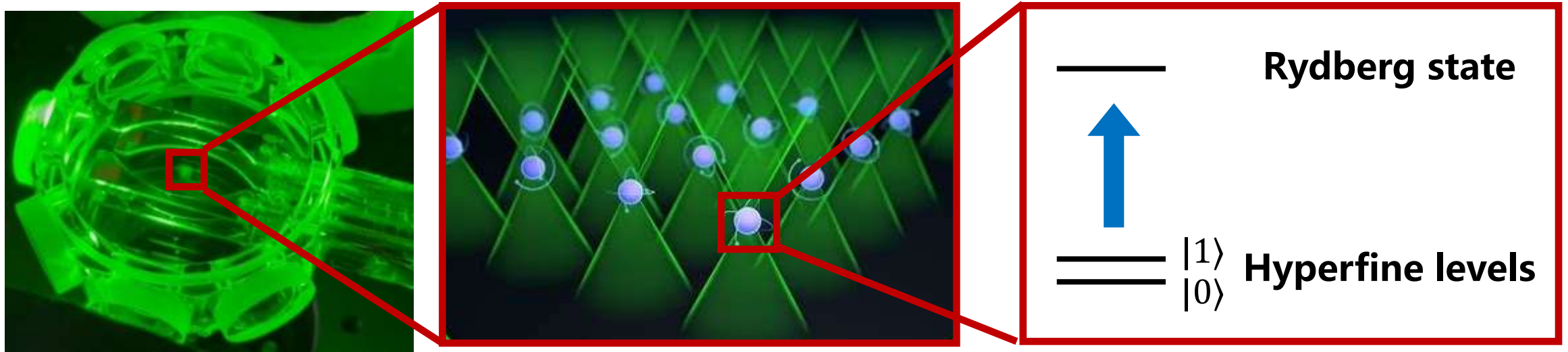
xrxie25@pku.stu.edu.cn

Frontiers Science Center for Nano-optoelectronics, School of Physics @ **Peking University**



Introduction to Yb atom array

Hardware Components

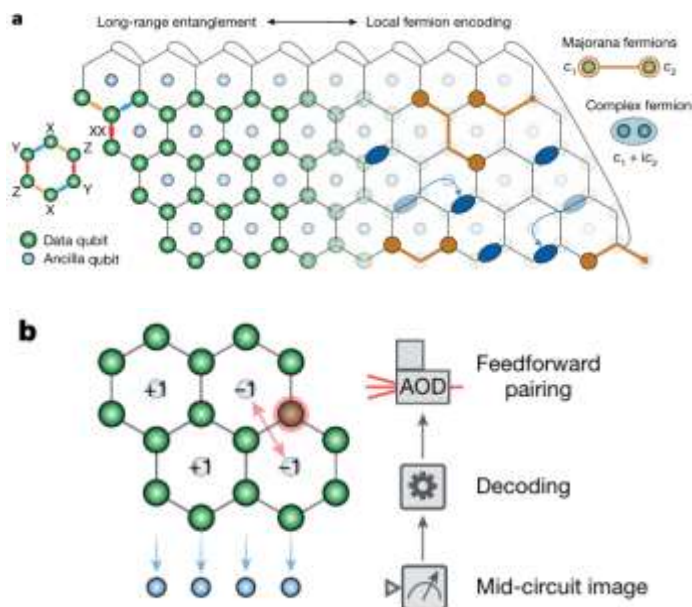


- ✓ Atoms trapped in tweezer arrays 👉 **Large scale**
- ✓ Qubits defined in hyperfine levels 👉 **Long coherence time**
- ✓ Quantum gates via Rydberg interaction 👉 **Strong interaction**

Selective Results

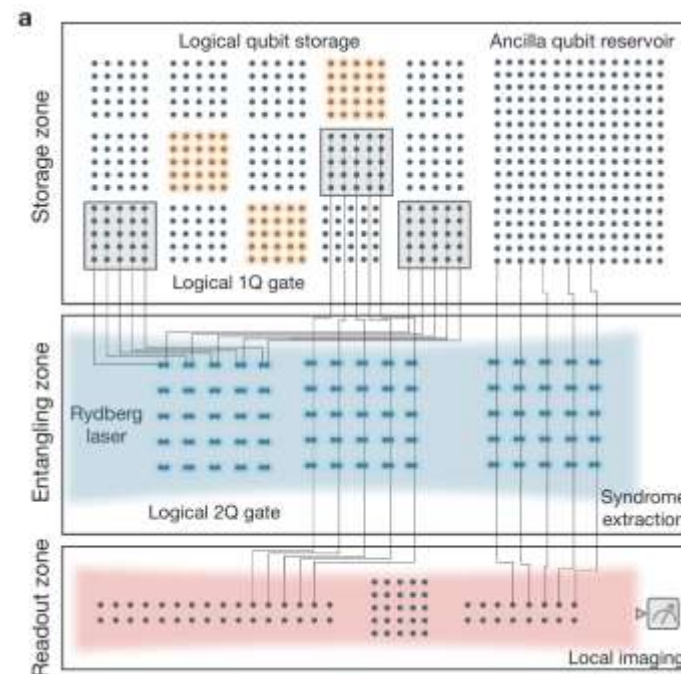
A **powerful** and **versatile** platform for quantum physics research.

Quantum Simulation



Kitaev Honeycomb^[1]

Quantum Computation



48 Logic Qubit^[2]

Quantum Metrology



Clock stability^[3]:
 $5.2 \times 10^{-17} / \sqrt{\tau}$

- [1] Evered, S.J., Kalinowski, M., Geim, A.A. et al. Nature (2025)
- [2] Bluvstein, D., Evered, S.J., Geim, A.A. et al. Nature (2024)
- [3] Young, A.W., Eckner, W.J., Milner, W.R. et al. Nature (2020).

Akali vs Akali-earth(-like)

The periodic table shows the following elements highlighted:

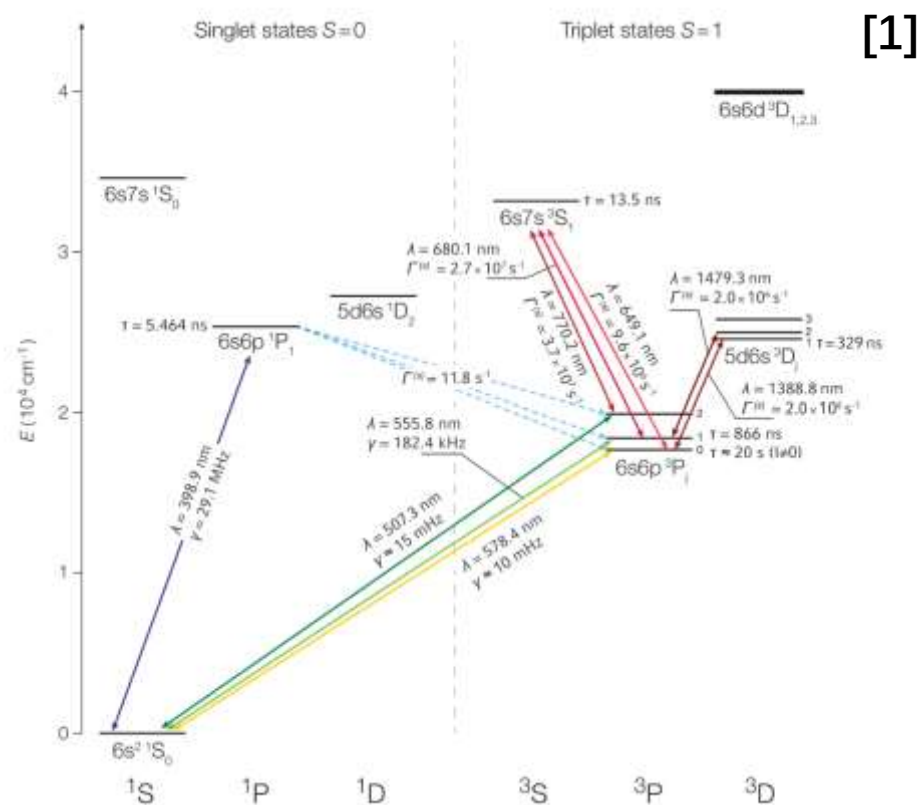
- Alkali metals (Group 1, blue box):** H, Li, Na, K, Rb, Cs, Fr.
- Alkaline-earth metals (Group 2, red box):** Be, Mg, Ca, Sr, Ba, Ra.
- Lanthanides and Actinides (f-block, red box):** Dy, Ho, Er, Tm, Yb (Lanthanides); Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No (Actinides).

The d-block is labeled between groups 3 and 10, and the f-block is labeled to the left of the lanthanide and actinide series.

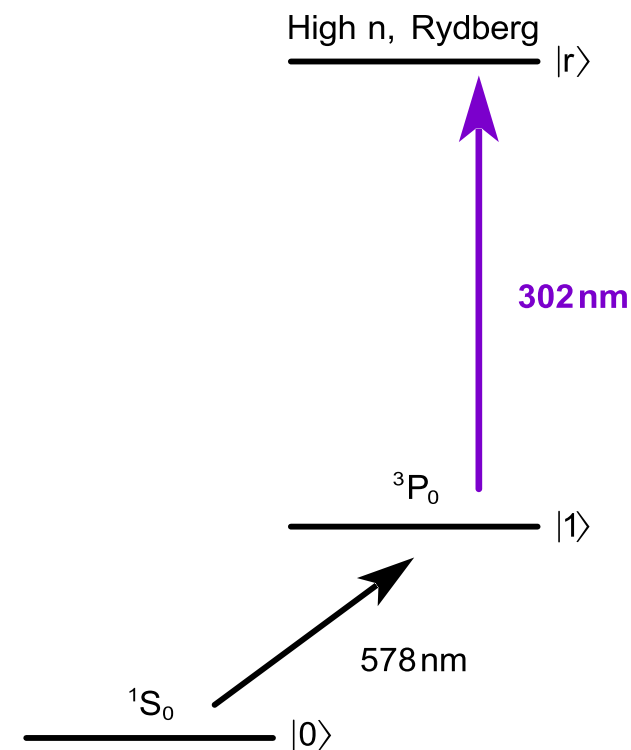
- Alkali atoms: One valence electron, **easy to control**
- Alkaline-earth(-like) atoms: Two valence electrons, **versatile in manipulation**



Why Ytterbium?



- ✓ Two stage cooling:
 - 399 pre cooling + 556 narrow line cooling $\rightarrow$ 5uK
 - No needs for sub-doppler cooling

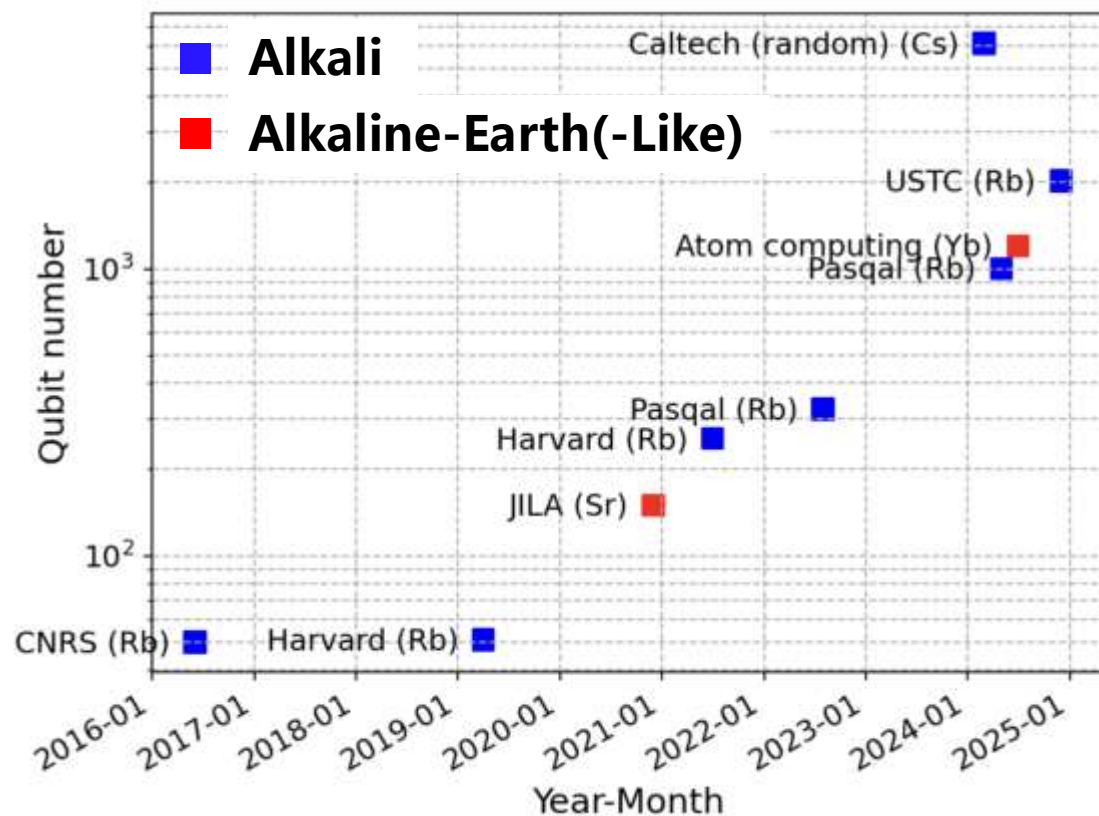


- ✓ Direct excitation $3P_0 \rightarrow$ Rydberg state at 302nm for two qubit gate.

[1] Kleinert M, et al. Phys. Rev. A (2016)

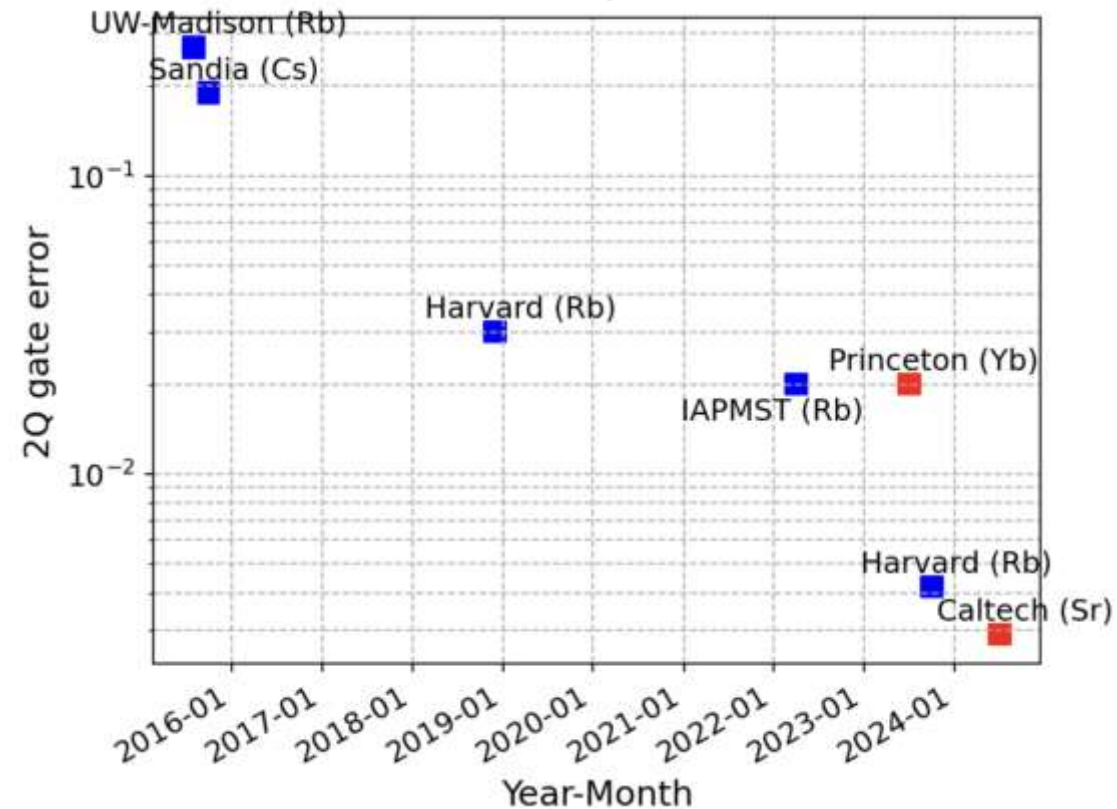
Development

Scaling Up!



Random: 6100; Defect-Free: 2024

Fidelity Up!

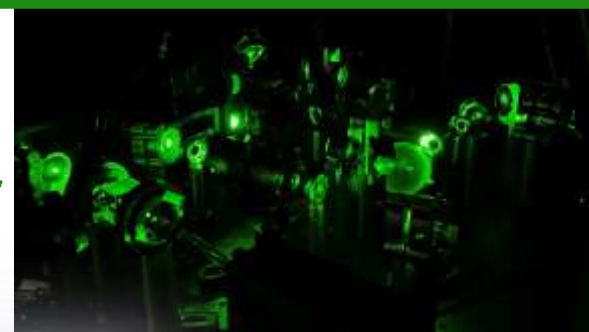


Two-qubit-gate fidelity: 99.71(5)%

Large scale Yb-174 atom array

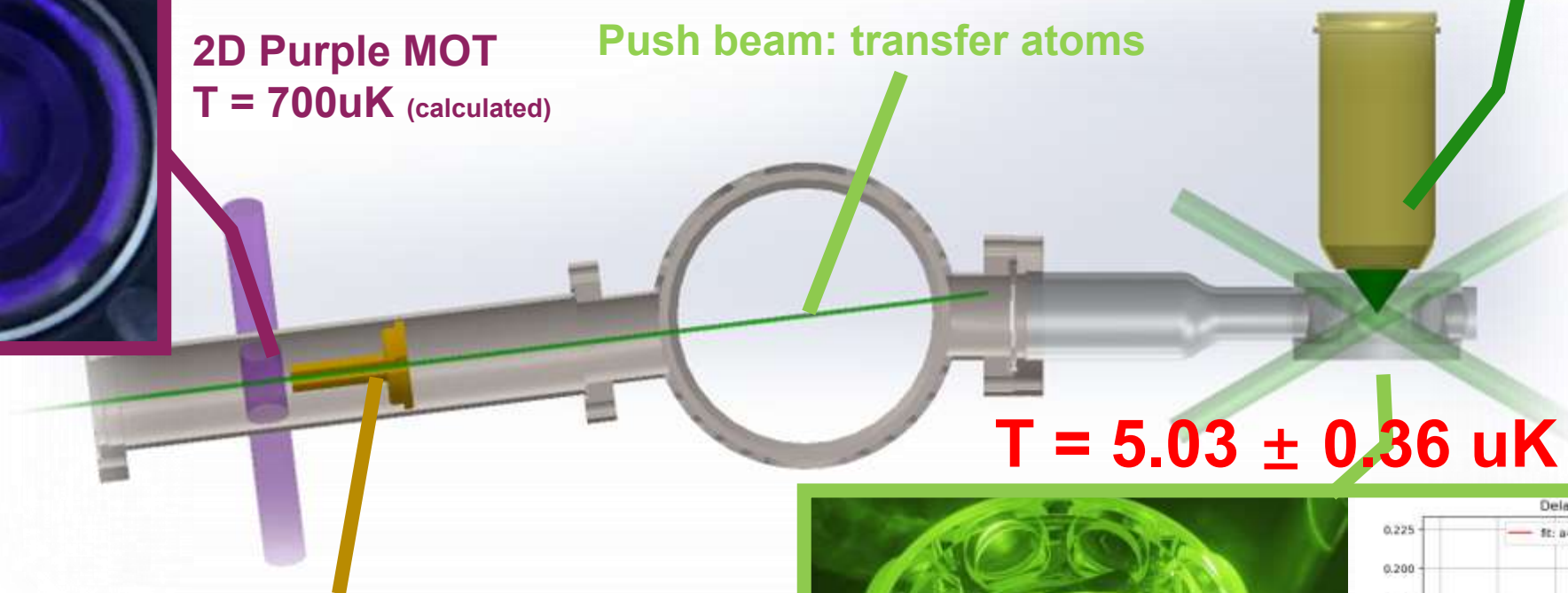
Two-stage laser cooling

High power
MAGIC tweezer



2D Purple MOT
 $T = 700\mu\text{K}$ (calculated)

Push beam: transfer atoms

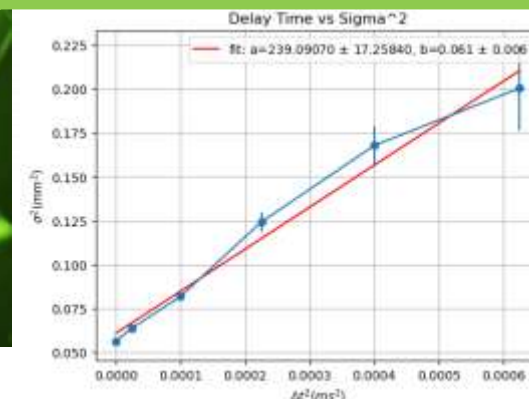


$T = 5.03 \pm 0.36 \mu\text{K}$

Differential Pumping
Tube
 $2\text{E-}8 \text{ Pa} \rightarrow 4\text{E-}9\text{Pa}$



3D Green MOT



Optical Tweezer: static

Spatial light modulator (SLM)

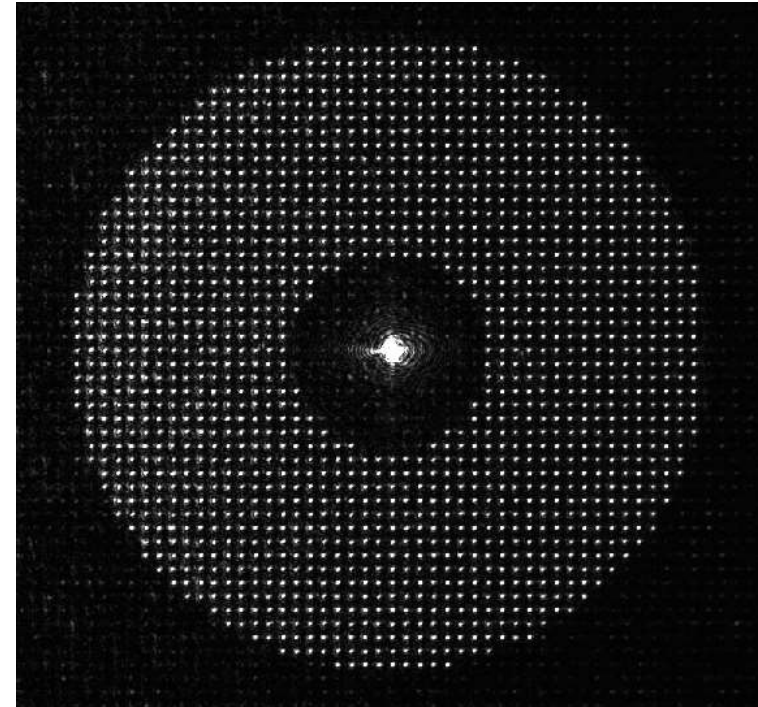


- Voltage control
- Pixelated array
- Programmable phase modulation

✓ **Large scale ~ 2000**

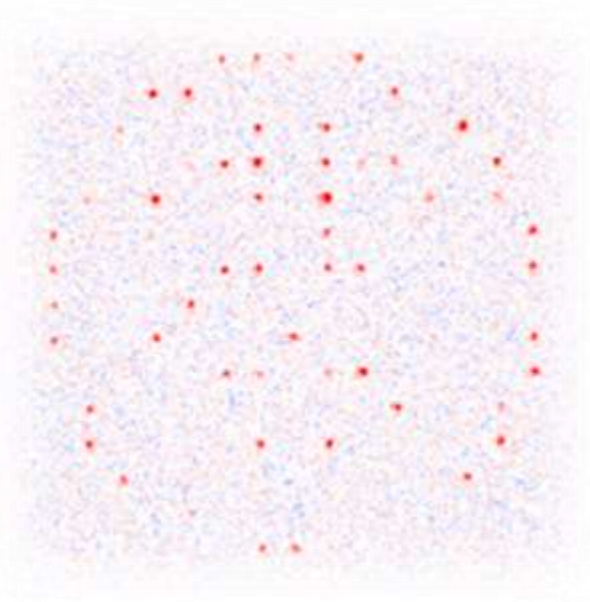
Low Frame rates ~ 60Hz

- To generate **static traps**



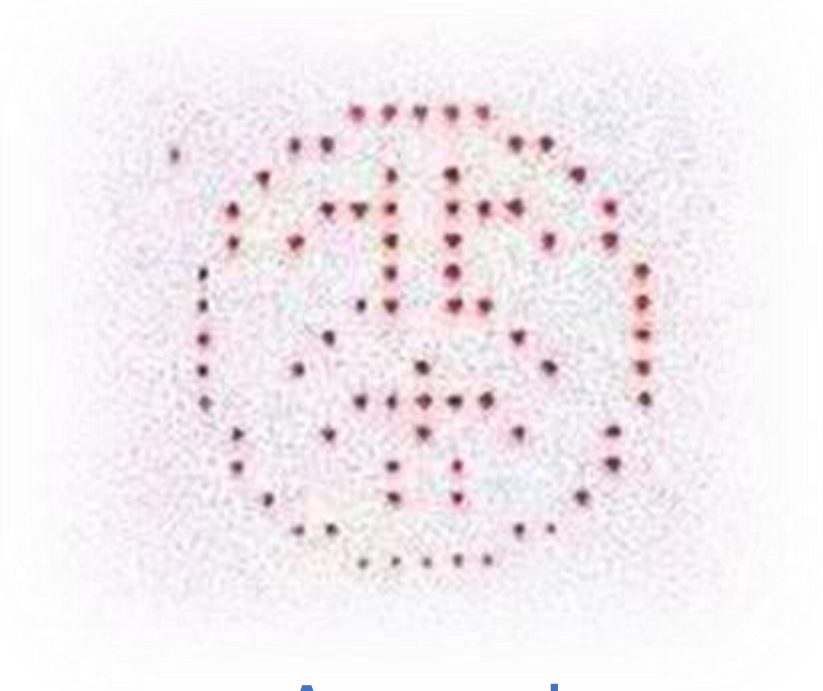
Large scale 2D atom array

Badge: Peking University



Single-Shot

Loading efficiency (randomly) $\sim 50\%$

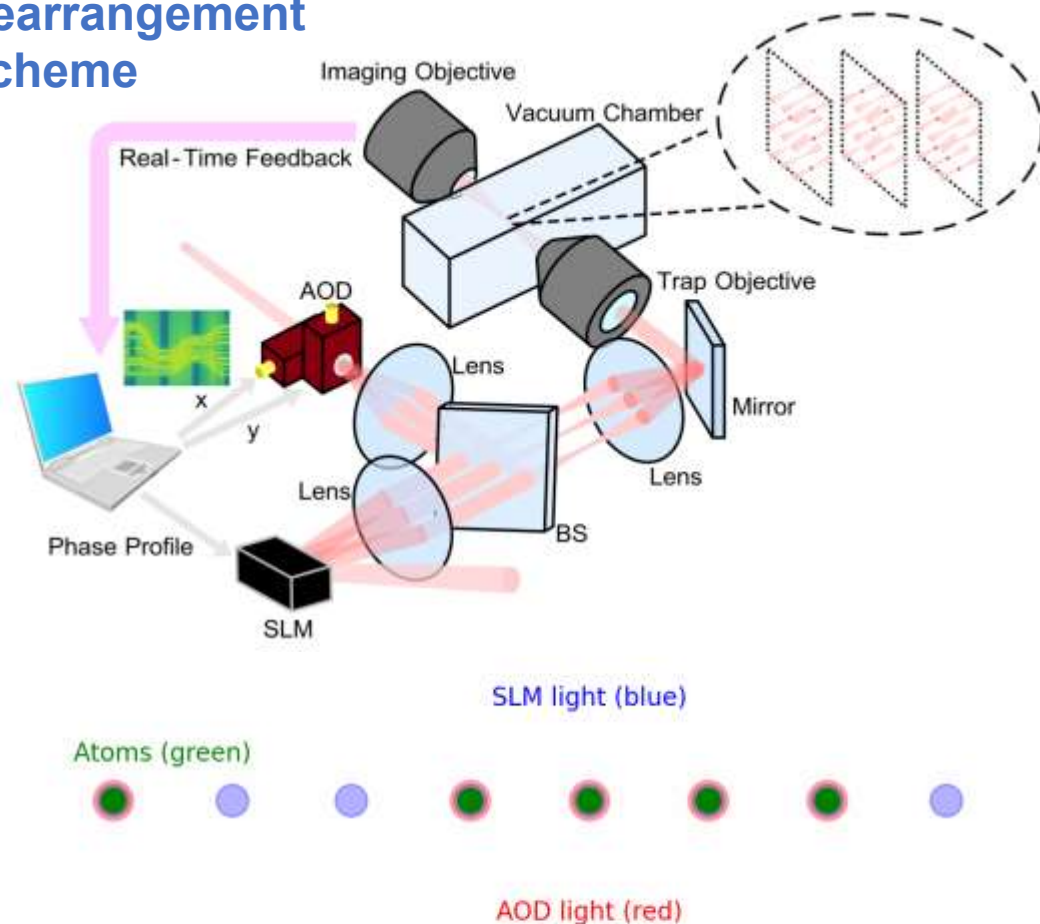


Averaged

Over 100 atoms

Rearrangement

Rearrangement Scheme



SLM + AOD towards large-scale programmable atoms array !!!

Acoustic optical deflector (AOD)



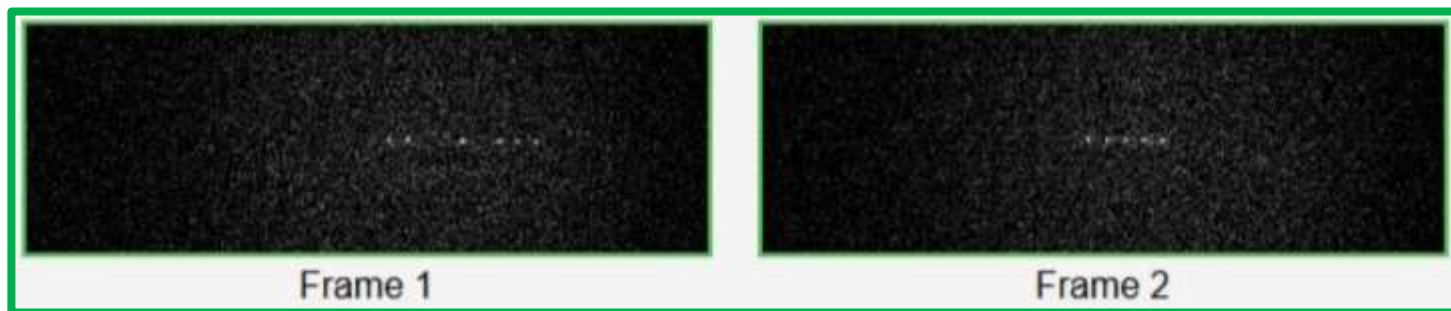
- AWG driving
- Laser beam deflection
- High speed scanning

Small scale ~ 100

✓ High Frame rates $\sim 10^2$ kHz

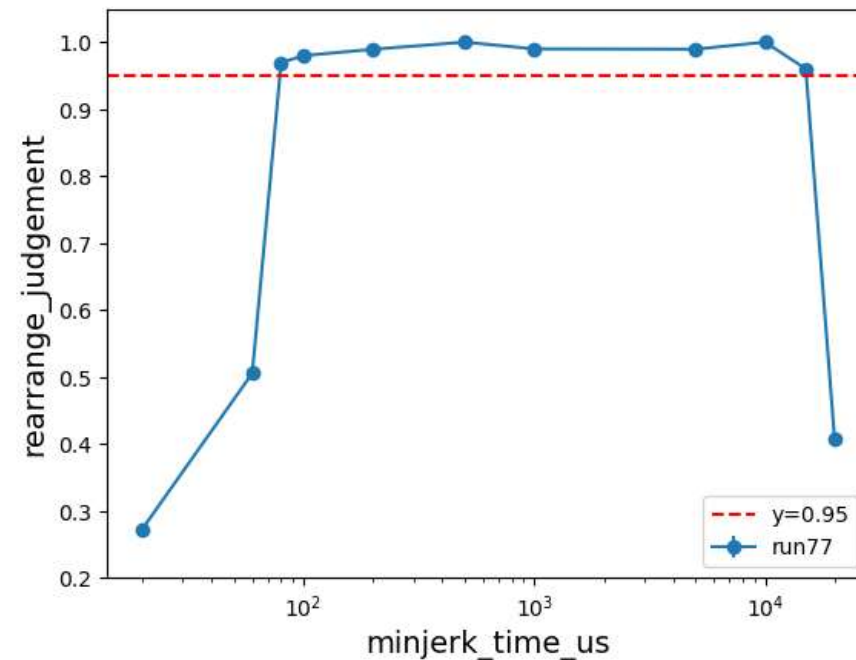
- To generate **movable tweezers**

High fidelity 1D rearrangement



11 tweezer (~50% loading efficiency)

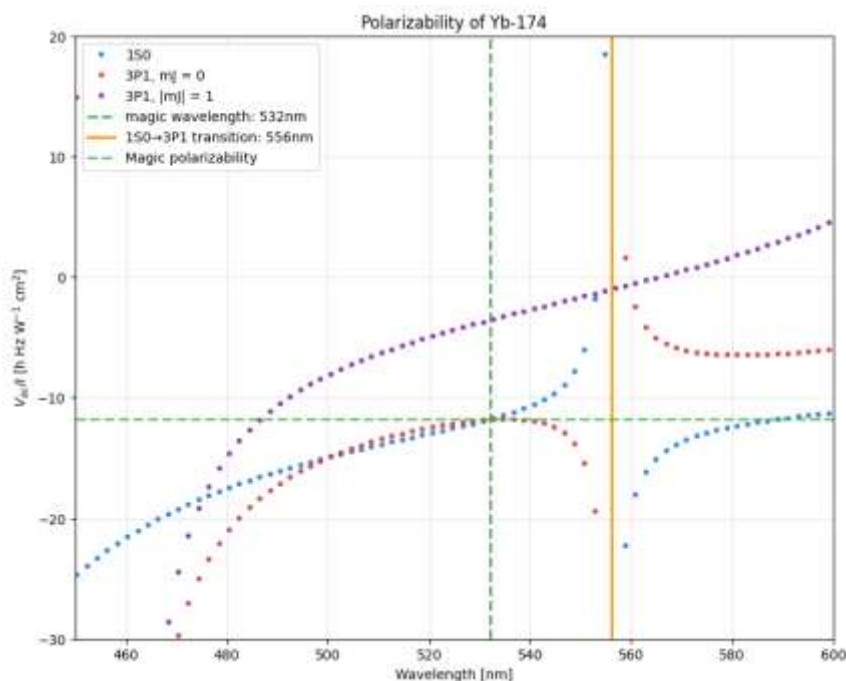
-> **defect-free array of 5 atoms**



Optimized Rearranged fidelity **over 99.9%**;
Wide time scale fidelity **over 95%**.

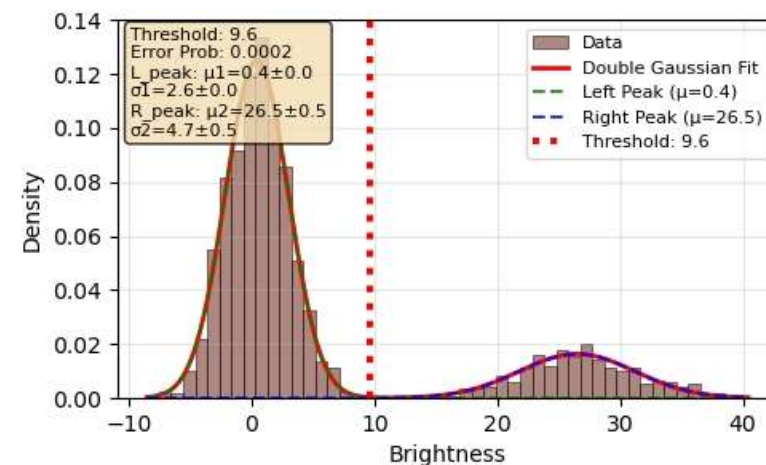
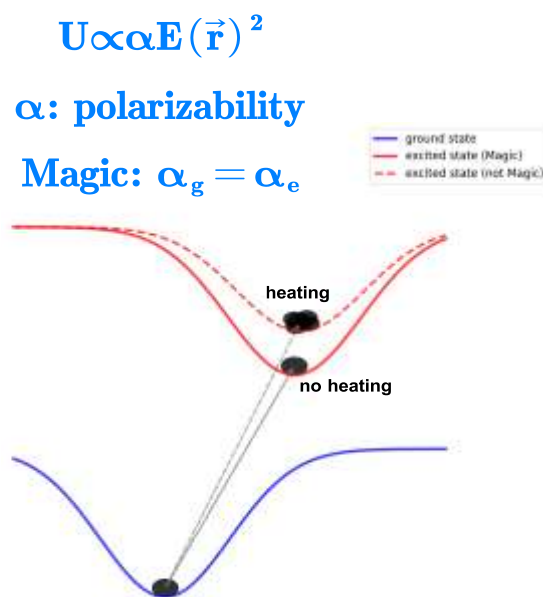
High fidelity imaging

Magic wavelength

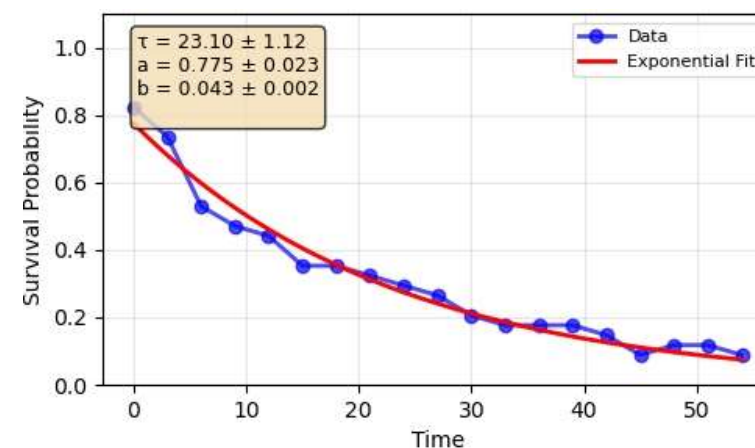


Yb-174 ($I=0$): $1S_0 \rightarrow 3P_1(m_j=0)$ is magic at 532nm 🙌 no heating

532 is easy to build: commercially available 30W laser



infidelity under 1E-3



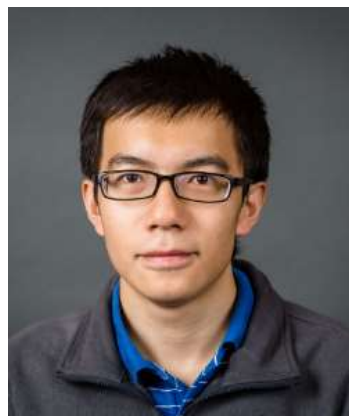
Imaging lifetime over 20s

Outlook

- Trapping over **2000 atoms**, largest alkaline-earth(-like) atom array
- **Yb-174 quantum gate scheme:**
 - ✓ 1S_0 & 3P_0 encode qubit
 - ✓ Single qubit gate: clock laser at **578 nm**
 - ✓ Two qubit gate: **302nm** single-photon Rydberg

Acknowledgement

Laboratory of Rydberg Atoms and Quantum Science @ PKU



PI: Pai Peng



Research Fellow:
Sheng Zhang

Experiment Setup:



Li Zhou



Jiawen Zhu



Chenyang Jiang

AOD:



Zhuoli Ding



Juxian Li

SLM:



Fan Wu



Rongpei Zhu



Changfen Chen

PhD & Postdoc available! Contact pengpai@pku.edu.cn

Q & A

Thanks

